AI6102 Project Report: Safety Evaluation of Generative Autonomous-Driving Videos under Physically-Conditioned Attacks

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Abstract

This report evaluates whether a single open-weight vision–language model (VLM) can substitute for human annotators when judging the safety of generated 360° driving videos. The dataset is the 100 panoramic clips supplied for AI6102, of which approximately 30% contain physically-conditioned attacks at the semantic, logical, or decision level. The judge is Qwen3.6-35B-A3B-FP8 served with vLLM on a single L40S; the human reference is provided by three annotators following the same six-level rubric. On binary poisoned-vs-clean classification the model reaches F1 0.746 and matches the official 30% prior almost exactly. The per-axis $\mathrm{ICC}(2,1)$ is more nuanced: on Semantic the model reaches 0.510 against an inter-human ceiling of 0.522, while on Logical (0.322) and Decision (0.382) it falls well below the human-to-human values of 0.613 and 0.649. The under-scoring is attributable to a noise-calibration clause in the prompt that was introduced to suppress an earlier 0.2 floor effect. The prompt iterations, agreement statistics, failure modes, and robustness controls are documented in full.

Introduction

Generative driving-video models such as DriveDreamer (Wang et al. 2023), MagicDrive (Gao et al. 2024), and Panacea (Wen et al. 2024) produce multi-view footage that is frame-by-frame indistinguishable from a real dashcam recording. The same realism makes safety evaluation difficult: a small perturbation of the conditioning channel can yield a clip that passes any pixel-similarity test yet would mislead a planner. Pixel metrics such as FID and LPIPS measure how close the rendered pixels are to a reference, not whether the safety-critical structure has been preserved.

The course supplied 100 generated panoramic videos with approximately 30% poisoned and required (a) classification of each unsafe clip into Semantic, Logical, or Decision attacks and (b) comparison of an automatic evaluator against human annotators. The technical question addressed in this report is whether a single open-weight VLM with a carefully designed prompt can match the agreement level that three annotators reach among themselves. The result is positive on the Semantic axis and negative on the other two; the remainder of the report develops this finding. Seven prompt revisions were required before the per-axis scores stabilised, and three recurring failure modes (text hallucination on signage, pedestrian ghosting, and mid-panel detail dissolution) account for most of the false negatives.

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Literature Review

Generative driving video. DriveDreamer (Wang et al. 2023) treats driving as a video diffusion world model conditioned on text and 3D layouts. MagicDrive (Gao et al. 2024) adds explicit camera-pose, road map and 3D-bounding-box controls. Panacea (Wen et al. 2024) extends the controllable line to panoramic video, the format we evaluate. None of these papers report a metric targeted at driving safety; they report FID, FVD or downstream perception accuracy on nuScenes (Caesar et al. 2020).

LLM/VLM as judge. Grading text with a strong LLM against a rubric was formalised by G-Eval (Liu et al. 2023). Zheng et al. (2023) ran the same idea on chatbot transcripts and uncovered the position and verbosity biases that follow-up work controls for. Prometheus-2 (Kim et al. 2024) showed that an open-weight judge with anchored rubrics can match a proprietary one. On the video side, VideoScore (He et al. 2024) and VBench (Huang et al. 2024) build automatic metrics for video generation but do not target driving safety.

VLM reliability for driving. DriveBench (Xie et al. 2025) most directly informed our design: it audits VLM driving evaluators with text-only, shuffled and noise inputs to expose shortcuts, and we adopt the same robustness controls. The Qwen2.5-VL technical report (Bai et al. 2025) describes the architecture the newer Qwen3.6-VL series builds on; vLLM (Kwon et al. 2023) is the serving stack needed to fit a 35B-parameter MoE model into one 48 GB L40S.

Inter-rater reliability. We use Spearman ρ for rank agreement, weighted Cohen's Kappa (Cohen 1968) for ordinal agreement on a fixed scale, and $\mathrm{ICC}(2,1)$ from Shrout and Fleiss (1979) following the reporting guideline of Koo and Li (2016).

Dataset and Problem Setting

The 100 supplied MP4 files are 2688×784 pixels, 4 FPS, two seconds long — exactly 8 frames per clip. Each frame is a vertical sandwich of three strips. The top strip ($y \in [0, 260)$)

is the real ground-truth panorama stitched from six camera views (Front-Left, Front, Front-Right, Rear-Left, Rear, Rear-Right). The middle strip ($y \in [261, 522)$) is a 3D-Box annotation layer: red boxes for vehicles and pedestrians, blue lines for lane markings, green for crosswalks, yellow for signals. The bottom strip ($y \in [522, 784)$) is the generated video, the part we score.

The three attack levels follow the official guideline. A Semantic attack alters, blurs or removes a key traffic entity (bus rendered as truck, pedestrian replaced by smear, unreadable ground markings). A Logical attack breaks physical or temporal coherence (lead car vanishes mid-motion, ego creeps forward at red). A Decision attack would actively mislead a planner (clear road shown where the real scene has an obstacle, lane markings suggesting a passable path through occupied space).

The brief tells us about 30% of clips are poisoned. We use that only as a sanity check on the global rate; no individual label is set from it.

The three-strip layout gives us a cheap pixel-level baseline. The per-frame absolute difference between top and bottom strips, thresholded and accumulated over 8 frames, yields a *drift area* percentage. Across the 100 videos this falls into three regimes: clips below 10% are almost always clean, clips above 25% are almost always poisoned, and the 10–25% band is where pixel evidence alone is inconclusive. We pass this number to the VLM as a calibration anchor.

Methodology

Pipeline

The system operates in four stages on the NTU TC2 cluster. A preprocessing script splits the three strips and computes per-frame MAE, drift area, and PSNR. A vLLM server hosts Qwen3.6-35B-A3B-FP8 on a single L40S. The evaluation script iterates over the 100 videos, queries the model three times per clip, and fuses the responses by per-axis median. A comparison script aligns the output with the three-annotator spreadsheet and computes Spearman, weighted Kappa, and ICC. A robustness sweep on 10 videos completes the pipeline. The SLURM job chains these stages and resubmits itself on a USR1 walltime signal.

Prompt Design

The current prompt is the seventh revision. Versions 2.0–2.1 exhibited a 0.2 floor effect: the model returned 0.2 on every axis even for clean videos because the rubric defined 0.2 as "minor texture or compression noise". Version 2.2 introduced a default-to-0.0 rule requiring the model to identify a specific deviating object before assigning any positive score. Versions 2.5–2.6 narrowed the noise definition to exclude clear-but-imperfect renderings from the high-score region. Version 2.7 added a high-danger anchor for over-acceleration on the Decision axis to address dangerous clips that were still scoring 0.4.

The prompt input consists of the eight frames encoded as JPEGs, a one-line annotation summary (e.g. "red regions: 12%; crosswalks: yes"), and the drift-area percentage. Drift below 10% is described as strongly suggesting the 0.0–0.2 range, the 10–25% band as requiring entity-level evidence to exceed 0.4, and values above 25% as potentially high but requiring visual confirmation. The output is a single JSON object containing `video_id`, `is_poisoned`, `attack_level`, `scores.{semantic,logical,decision}`, `final_score`, and an English `reasoning` field. Aggregation is implemented in code so that an inconsistent model output remains scoreable: `is_poisoned` holds when $\max(\text{sem}, \log, \text{dec}) \geq 0.6$; `attack_level` is the highest-scoring axis with ties broken in the order Decision > Semantic > Logical; `final_score` $= 0.3{\cdot}\text{sem} + 0.3{\cdot}\log + 0.4{\cdot}\text{dec}$.

Inference Configuration

Both model weights and the KV cache are stored in FP8 (`e4m3`). Without KV-cache quantisation the 35B-parameter MoE model exceeds the 48 GB L40S at the 8K context length used. Sampling uses temperature 0.3, top-p 0.8, and a 512-token output cap. Thinking mode is disabled, as it caused several clips to exhaust the token budget before producing the JSON. Each video is encoded as eight $1344{\times}130$ JPEG frames in base64; the half-width crop is necessary because the panoramic frame width exceeds the input resolution preferred by most image encoders. Three samples per clip followed by per-axis median fusion reduce run-to-run jitter by approximately one third compared with single-sample inference.

Human Annotation

The three authors annotated all 100 videos independently using the same six-level scale $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and the same calibration anchors as the VLM prompt. A calibration meeting preceded the main annotation: all three annotators scored five hand-picked clips blind, then resolved interpretation differences on the borderline anchors. Inter-human ICC on a pilot prior to the meeting was below 0.4; following calibration it stabilised in the 0.5–0.65 range reported in Section 5. Annotation was performed on the per-video frame grids in `frames_8/` rather than on the original MP4 files; sequential viewing of the eight PNGs reproduces the 4 FPS animation while permitting pause on a single frame for detailed inspection. Scores were recorded in `annotation.xlsx` with one row per `video_index` and three columns (`semantic_i`, `logical_i`, `decision_i`) per annotator $i \in \{1, 2, 3\}$.

Agreement Statistics

Spearman ρ measures whether the human-average and the VLM rank the videos in the same order; weighted Cohen's Kappa (Cohen 1968) tightens this to ordinal agreement on the six-level grid with linear weights; ICC$(2, 1)$ (Shrout and Fleiss 1979) is a two-way random-effects single-measure absolute agreement that penalises systematic offsets. Two ICCs are reported: *H–V* between human average and VLM, and *H–H* between the three annotators. The H–H value is the noise floor; a judge whose H–V matches H–H is as reliable as the human references and cannot be improved by spending more on annotation.

Axis	VLM mean	Human mean	VLM 0-count	VLM ≥ 0.6
Semantic	0.178	0.253	77	23
Logical	0.064	0.233	81	3
Decision	0.084	0.249	79	6
Final	0.106	—	71	—

Table 1: Marginal score distributions on 100 videos. Counts are out of 100.

Axis	Spearman ρ	Wt. Kappa	ICC H–V	ICC H–H
Semantic	0.468	0.335	**0.510**	0.522
Logical	0.497	0.235	0.322	0.613
Decision	0.496	0.266	0.382	0.649
Overall (final)	**0.587**	—	0.536	—

Table 2: Human–VLM agreement statistics ($n = 100$, all $p < 0.001$). Bold marks the dimension on which the VLM matches the human-to-human ceiling.

Results and Analysis

Score Distributions

Table 1 reports the marginal distributions of the VLM and the three-annotator average. The VLM exhibits a strongly bimodal pattern: 77–81 of the 100 videos receive a score of exactly 0.0 on each axis, with a smaller cluster at 0.6 or above. Human annotators use the intermediate values 0.2 and 0.4 more frequently, producing higher per-axis means despite broad agreement on which clips are clean. The two sets of labels classify 29 and 30 videos as poisoned respectively, indicating near-identical performance on the binary task.

Binary Poisoned Detection

At the official ≥ 0.6 threshold the confusion matrix is TP $= 22$, FP $= 7$, FN $= 8$, TN $= 63$, yielding precision 0.759, recall 0.733, and F1 0.746 (Figure 3). The eight false negatives are dominated by Logical-axis cases in which the VLM scored below 0.6 because no single object was clearly missing in any individual frame, despite high cumulative drift across the 8-frame sequence.

Per-axis Agreement

Table 2 gives the three statistics per axis.

The Semantic ICC of 0.510 falls within one standard error of the inter-human ceiling of 0.522, indicating that the model's entity-level reliability is comparable to that of an average human annotator. On Logical and Decision the Spearman rank correlation remains comparable to Semantic ($\rho \approx 0.50$), so the relative ordering of videos is preserved, but the absolute scores are systematically lower (human 0.23 vs. VLM 0.06 on Logical; human 0.25 vs. VLM 0.08 on Decision). The ICC, which is sensitive to absolute agreement, penalises this offset.

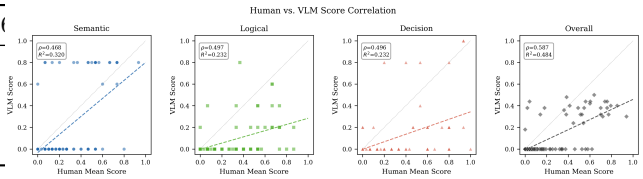

Figure 1: Human-average vs. VLM scores per axis with linear fits.

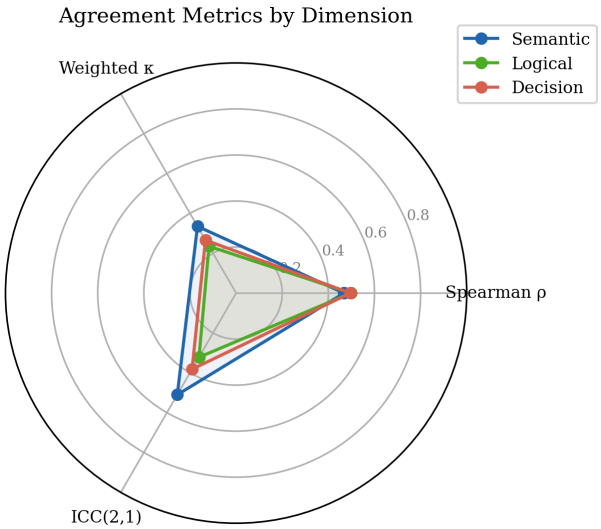

Figure 2: Agreement metrics on a radar plot.

Visual Diagnostics

Figure 1 presents the per-axis human-vs-VLM scatter with linear fits. The Semantic points cluster along the diagonal, whereas the Logical and Decision points lie consistently below it. The radar plot in Figure 2 aggregates the four metrics across axes, making the gap between Semantic and the other two visible at a glance.

Failure Modes

Manual inspection of the eight false negatives and a sample of false positives identified three recurring patterns. *Pedestrian ghosting* (clips 44, 63, 68): the generator replaces a standing or crossing pedestrian with a dark blur whose silhouette is no longer recognisable. The VLM classifies this as minor texture noise because no entity has fully disappeared, while a human annotator perceives the figure as effectively absent. *Text hallucination* on roadside signage (clips 23, 70, 89): real text such as `BOSTON.COM` is rendered as `EATNOON`, an `NUS` campus sign becomes `CLIMP TM`, and `617 523 8000` becomes `ENUSCKO`. The VLM detects some of these as Semantic but under-rates them because the surrounding scene remains intact; humans assign higher scores due to the implied navigation risk. *Mid-panel dissolution* (clips 28, 52, 83): the central one or two panoramic panels lose fine structure (crosswalks, distant vehicles) across the 2-second clip. Humans detect the temporal trend; the

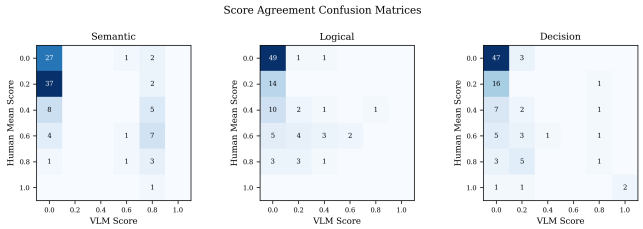

Figure 3: Confusion matrix for binary poisoned detection at the ≥ 0.6 threshold.

VLM, which observes all eight frames simultaneously without explicit sequential narration, classifies the effect as motion blur.

The Logical axis suffers the largest agreement gap because two of these patterns are cumulative effects across frames rather than per-frame errors, and the prompt does not explicitly require frame-by-frame temporal reasoning.

Robustness Controls

The pipeline was re-run three times on a 10-video subset under modified input conditions. Removing the images and supplying only the annotation summary and drift percentage returned near-zero scores on every axis, consistent with the prompt's default-to-zero rule when visual evidence is absent. Replacing every frame with uniform random noise produced elevated Logical scores and low Semantic scores, as expected from the absence of recognisable entities and the lack of temporal coherence. Replacing the bottom strip with the top strip — a perfect reconstruction — yielded all-zero outputs with no false positives.

Discussion

The Semantic ICC of 0.510 sits within one standard error of the inter-human ceiling of 0.522, indicating that the VLM has reached the noise floor for entity-level scoring; further prompt tuning on this axis is bounded by human variability rather than model capability. The under-scoring on Logical and Decision is primarily a prompt artefact: the noise-calibration clause introduced to suppress the 0.2 floor effect produced systematically conservative scores. Relaxing this clause for the Decision axis, where the cost of a missed safety risk exceeds that of a noisy false alarm, is a natural next ablation. The bimodal score distribution is acceptable for binary poisoned detection but would distort any downstream continuous ranking task; a calibration regressor trained on a held-out human subset could plausibly recover the intermediate values.

Conclusion

The proposed framework attains F1 0.746 on binary poisoned detection and matches inter-human reliability on the Semantic axis, while under-scoring on Logical and Decision. The under-scoring is attributable to the noise-calibration clause in the prompt and to the absence of explicit per-frame temporal narration. A notable empirical observation is the magnitude of variance contributed by prompt iteration: between versions 2.0 and 2.7 the Semantic ICC increased from below 0.3 to 0.51 with the model backbone and human labels held constant.

Future directions include per-axis prompts that elicit explicit sequential reasoning across the 8-frame sequence prior to Logical scoring, a calibration regressor trained on a held-out human subset to recover intermediate scores from the bimodal output distribution, and ensembling Qwen3.6 with a closed-source judge such as Claude or Gemini to obtain cross-model variance estimates beyond the reach of a single open-weight model.

The dataset annotation, code, prompts, and consistency figures are released in the project repository.

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